Assignment 2 – Prompt Engineering & Chunking Strategy

Page 1 – Task 1: Prompt Engineering

Objective: Test three prompts and observe how negative constraints change the response.

Experiment	Prompt / Constraints	Result
1. Basic	Explain Cloud Computing to a beginner.	General answer; simple but not strongly controlled.
2. Negative constraints	Explain Cloud Computing in simple English. **Do not use technical jargon, do not mention specific cloud providers, and do not exceed 50 words.**	Shorter and easier to understand; unwanted jargon/details are reduced.
3. Strong constraints	Explain Cloud Computing using a real-life example in 60–70 words. **Do not use acronyms, technical terms, bullet points, or complicated explanations.**	Most controlled output: length, example, style, and forbidden content are specified.

Sample Outputs

1: Cloud Computing means using computer services through the internet instead of relying only on your own device.

2: Cloud Computing means using computer services through the internet. Instead of storing everything on your own computer, you can use online resources to store files, run applications, and access information from different devices.

3: Cloud Computing is like renting a storage room instead of keeping everything in your home. You can store your files online and access them whenever you need them. For example, when you save photos online, you can open them from your phone or computer without keeping all the files on one device.

Conclusion: Negative constraints improved control over length, vocabulary, style, and unwanted content. The third prompt produced the most controlled response.

Assignment 2 – Prompt Engineering & Chunking Strategy

Page 2 – Task 2: Chunking Strategy

Objective: Compare fixed-size chunking with an alternative section-based strategy and test retrieval.

Original method – Fixed-size chunking: Text is split after a fixed number of characters. This can separate related sentences or parts of the same topic.

Alternative method – Section-based chunking: The document is split using headings, keeping each topic with its supporting sentences.

Chunk	Content
1 – Databases	A database is an organized collection of information that can be stored and managed efficiently.
2 – SQL	SQL is a language used to communicate with databases. It allows users to retrieve, insert, update, and delete data.
3 – Database Security	Database security protects stored information from unauthorized access and prevents data from being changed or deleted without permission.

Retrieval Test

Question: What can SQL be used for?

Fixed-size result: The answer may be incomplete if the sentence is split across chunks.

Section-based result: The complete SQL chunk is retrieved: “SQL is a language used to communicate with databases. It allows users to retrieve, insert, update, and delete data.”

Result: The alternative strategy preserved the complete topic context and successfully retrieved the information required to answer the query.

Conclusion: Section-based chunking is effective for structured documents because related information remains together, reducing context loss during retrieval.